Data-driven modelling (2)

Macroscopic modelling (direct demand modelling)

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Recap of four step transport model

- Prerequisites:
  - tessellation of space (e.g. postcodes, wards)
  - Demographic data

1. Trip generation (how many trips originate from each origin, O, and arrive in each destination, D).
2. Trip distribution (estimate O-D pairings)
3. Mode choice
4. Traffic assignment (route choice)

Four step transport model issues

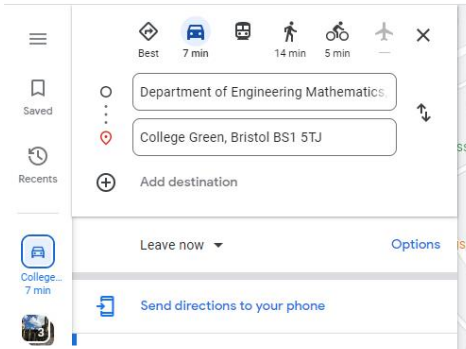
- Number of OD pairs can become difficult to manage:
  - E.g., when finer spatial scales are considered, and trips are shorter
  - For pedestrians: transport stops, schools, office buildings...

- Demand may change over short timescales:
  - Regular patterns: rush hour, end of the school day, and lunch time
  - Irregular patterns: delayed train arrival, protest march, street market

- … such dynamics can make alternatives that do not depend on the four steps and OD pairs attractive.

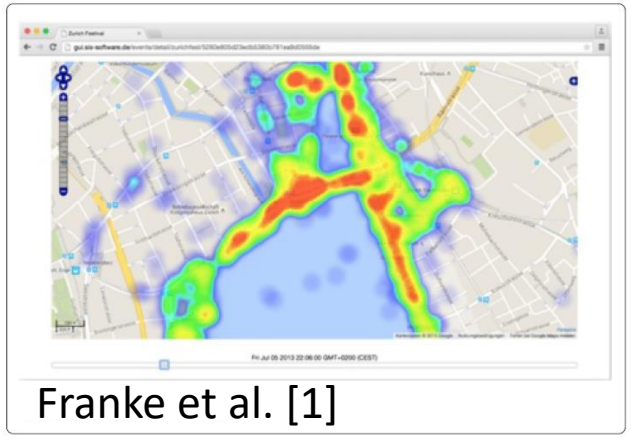
Alternative 1: direct measurements

- Obtain direct, live estimates of traffic Data from sat navs, mobile phones, WiFi, etc.
- Correct for sampling biases and estimate errors (if possible).
- Examples: Google Street Maps, App users at festivals, e.g. [1].



Google Street Maps





Franke et al. [1]

Direct measurements: issues

- Sampling issues (e.g., smartphone usage in children vs the elderly)

Berlin artist uses 99 phones to trick Google into traffic jam alert

Google Maps diverts road users after mistaking cartload of phones for huge traffic cluster

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Mon 3 Feb 2020 16.05 GMT



Google Maps Hacks by Simon Weckert.

A Berlin-based artist managed to create a traffic jam on one of the main bridges across the Spree with nothing but a handcart and 99 second-hand phones. But one other thing was unusual about the jam: it only existed on Google Maps.

Simon Weckert's artwork Google Maps Hacks involved the artist pulling a small red cart at walking pace down some of the main thoroughfares of

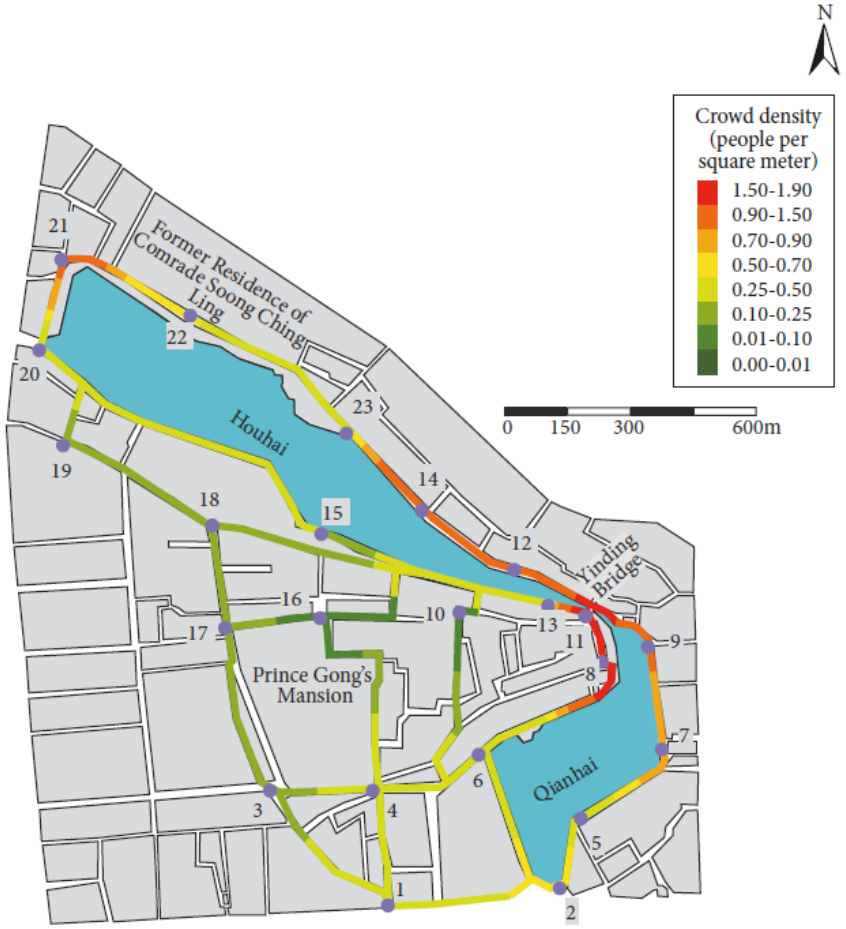
Direct measurements: issues

- Sampling issues (e.g., smartphone usage in children vs the elderly)
- Privacy issues (people should opt in)
- Data ownership issues (data often owned by companies, terms and prices may change at short notice, e.g., Twitter in 2023)
- Issues in sustaining long-term data collection (technology changes, people opt out of apps/services,…)

Alternative 2: direct demand modelling

- Use traffic observations at several distinct locations to fit/train models for predicting traffic* at other locations.
- Replace four steps model with one model for estimating traffic directly from built environment or sdociodemographic features.
- Approach 1: interpolation.
- Approach 2: regression.
- Typically, both approaches make use of representing transport infrastructure as spatial networks.

*here traffic can mean flow, density, counts, or similar.

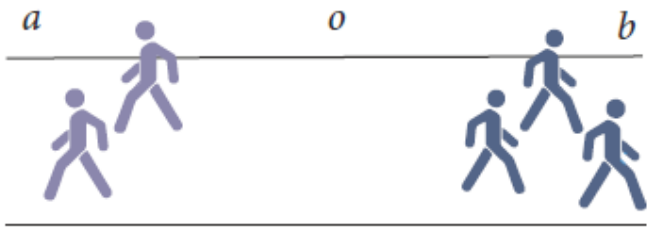


Wang et al. 2019 [2]

Direct demand modelling: interpolation

- Approach 1: Interpolation – assume traffic ρ_o at site o can be predicted as a function of traffic at other sites, ρ_i. Site o can be a road segment, or junction.
- Simple example: *Inverse Distance Weighting*, where weight depends on the distance between o and i, $D_{o,i}$:

$$\rho_o = \frac{\sum \rho_i / D_{o,i}^2}{\sum 1 / D_{o,i}^2}$$



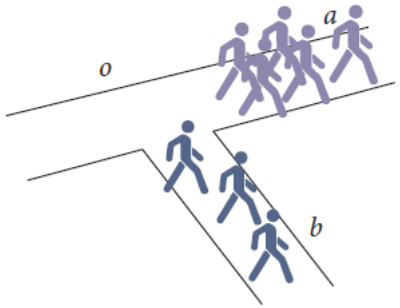
e.g. $$\rho_o = \frac{\frac{\rho_a}{D_{o,a}^2} + \frac{\rho_b}{D_{o,b}^2}}{\frac{1}{D_{o,a}^2} + \frac{1}{D_{o,b}^2}}$$

Wang et al. 2019 [2]

Direct demand modelling: interpolation

- Extensions: adjust distance calculations, obtaining re-weighted distances $D_{o,i}^{*2}$, to account for e.g., route preferences $P_{o,i}$ [2], or other route properties.





$$\rho_o = \frac{\sum \rho_i / D_{o,i}^{*2}}{\sum 1/D_{o,i}^{*2}}$$

$$D_{o,i}^{*2} = D_{o,i}^{2} / P_{o,i}$$

Wang et al. 2019 [2]

- Extensions: consider time-dependencies in route properties/preferences, $x_{o,i}$, e.g. $D_{o,i}^{*2} = f(D_{o,i}^{*2}, x_{o,i})$, for some function $f(\quad)$.
- Spatial statistics models, e.g. to calculate uncertainty in predictions (Kriging etc.).
- Note that distances can be defined in different ways, not just length in metres.

Direct demand modelling: regression

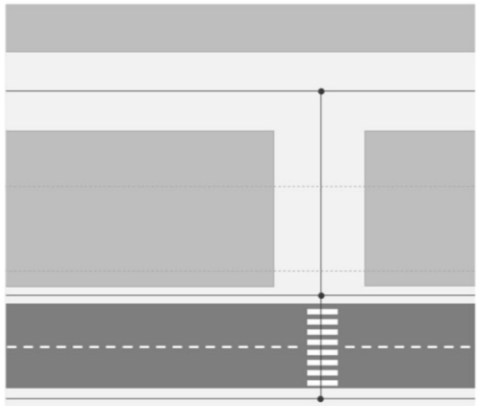
- Approach 2: Regression - assume traffic ρ_o at site o can be predicted as a function of variables y_o that characterise the location (can also include traffic at other sites, ρ_i, and previously observed traffic at o).

- Rationale: there are fundamental properties of the built environment, and especially the road/path network that allow us to predict traffic.

- (Simplistic) example: $\rho_o = \underbrace{\#shops_o + \#cafes_o}_{\text{predictors/covariates } y_o} + \epsilon$ ← *noise term*, a random variable

… sensible models consider the road network…

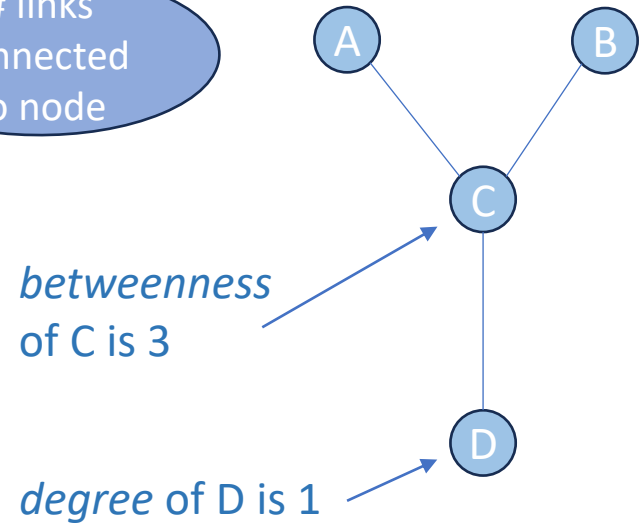
Topological features - principle


Cooper et al. 2021 [3]

- Can represent roads as a network consisting of nodes and links (not as easy as it seems).
- Network theory suggests *centrality measures* for measuring the importance of nodes, e.g. *degree* and *betweenness*.

shortest paths between all pairs of nodes that pass through a node

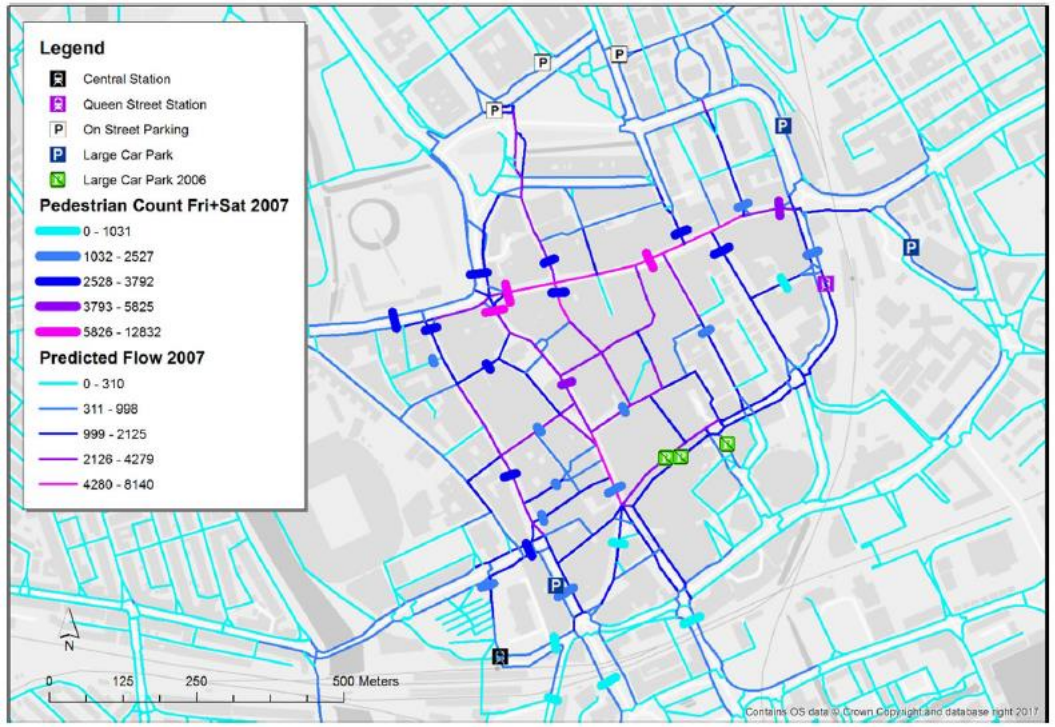
links connected to node



- If movement were random, nodes with high centrality measures would see the most traffic.
- Direct demand models use regression to test the predictive power of centrality measures.

Topological features – example [3]

Pedestrian traffic in Cardiff city centre



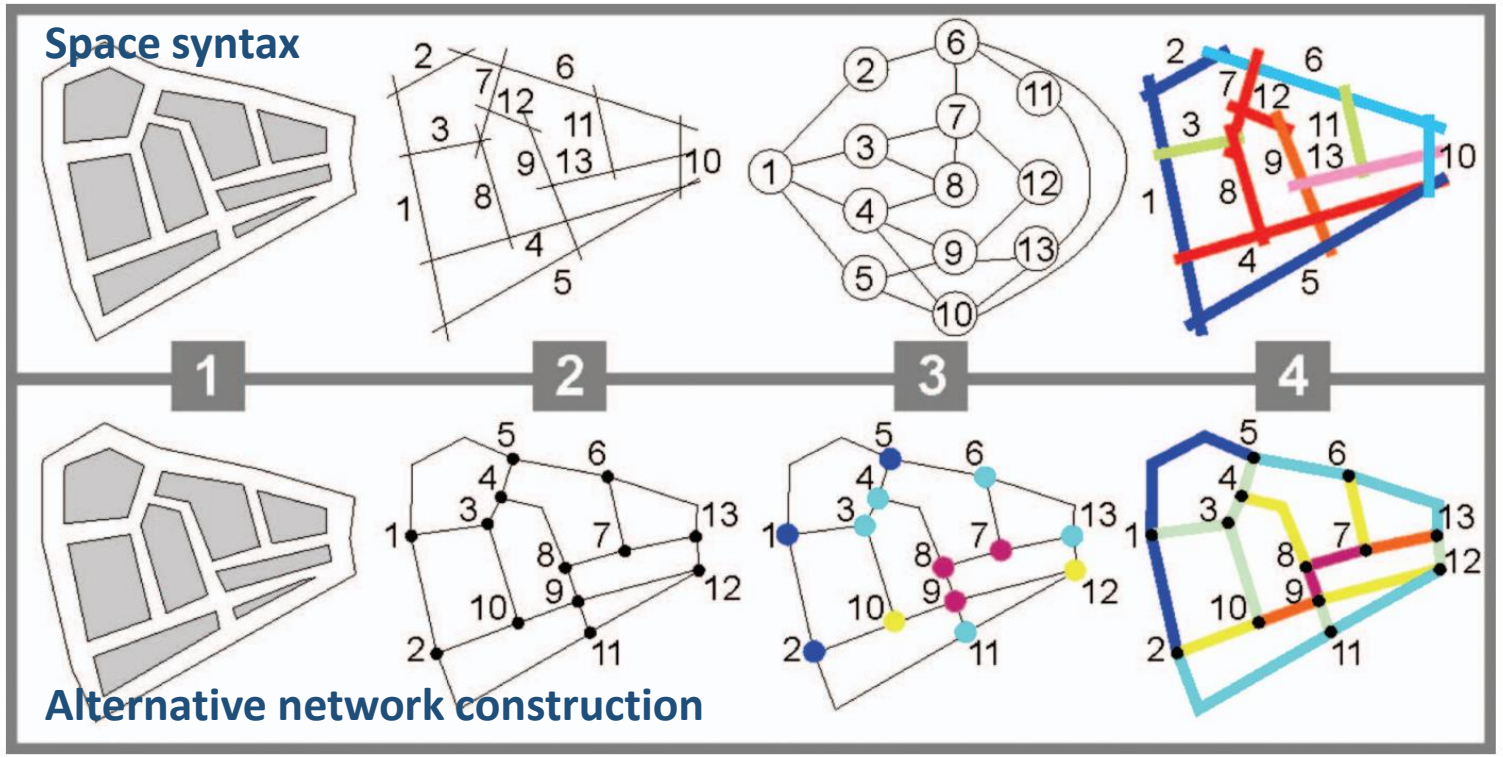
Cooper et al. 2021 [3]

- Develop new betweenness measures (parameterised):
  - Shortest paths defined using distance metric that considers path length and angle of turns.
  - Instead of all vertex pairs, consider paths between public transport, parking, and retail floor space.
- Consider multiple models, each with multiple covariates, including multiple betweenness measures.
- Use *ridge regression* to deal with correlations across different covariates.

Space syntax* - principle

* Pioneered by Hillier and Hanson, e.g. [4]

- Basic spatial unit is the line of sight.
- Street pattern is thus transformed into *axial map* (2).
- From this get the *connectivity graph* (3) (dual graph)
- From (3), compute the *integration index* (in practice this is a differently normalised version of the *closeness centrality*)



Crucitti et al. 2006

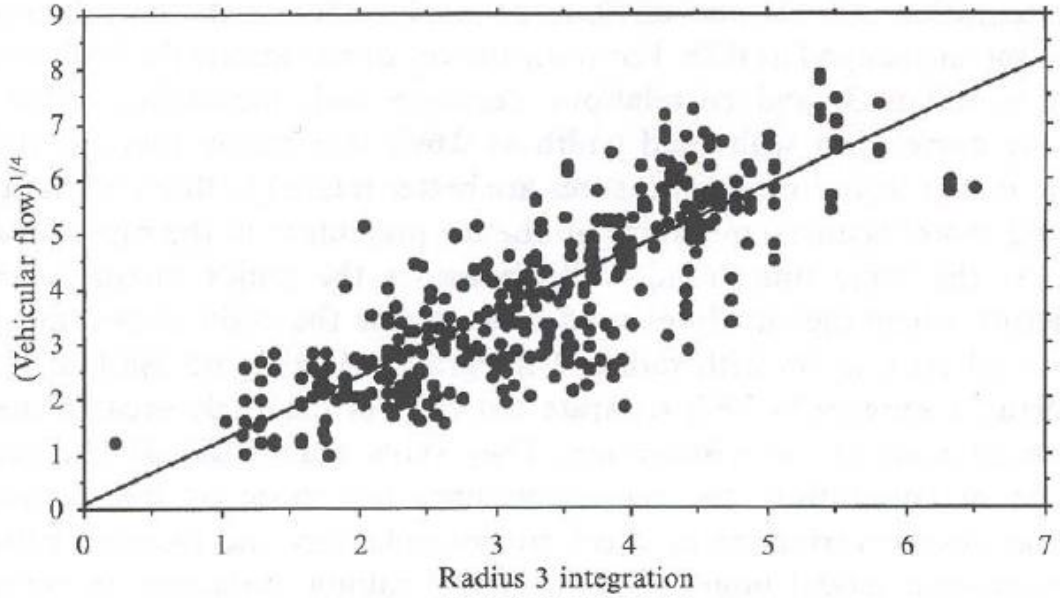
Reciprocal of sum of shortest path lengths connecting focal node to others. Higher values imply a more central position

Space syntax – example [4]



Radius 3 integration index map

Traffic in Greater London (1998)

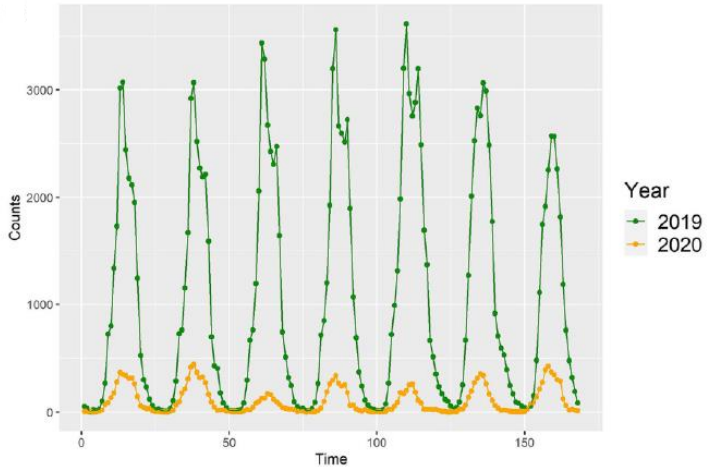


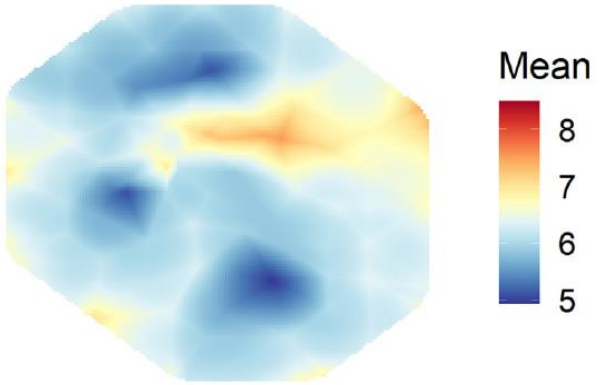
Correlation of measured all-day average hourly vehicle flow with radius 3 integration index: 0.82

Note the ¼ exponent on the y-axis!

Other features

- Built environment and other environmental covariates: shops, restaurants, bars, offices, green space, proximity to public transport, schools, vehicle traffic, tourist sites…
- Transport data often displays regular patterns, e.g. weekends, national holidays, day/night…
- Data-based: spatial and temporal (auto-) correlations, e.g. [5]. *Note the link to interpolation and see also last lecture for this section.*

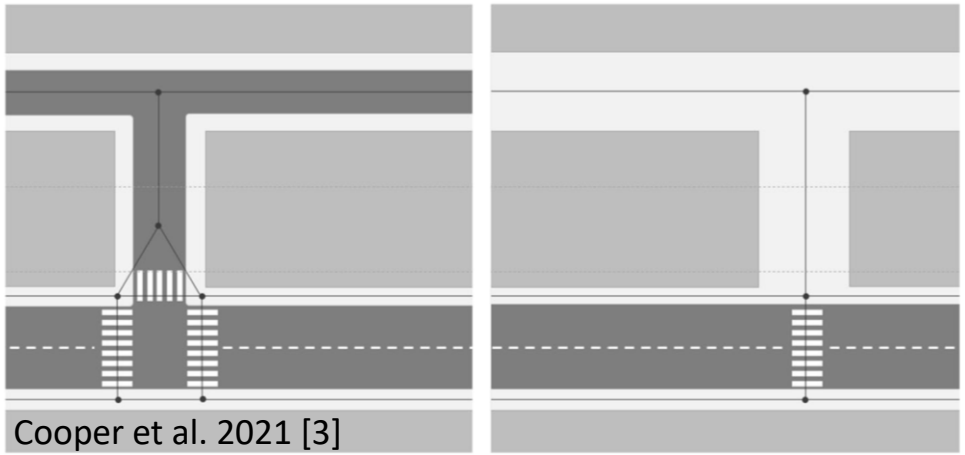




Direct demand modelling: issues

- Data limitations, e.g., locations with available data can induce biases (often only busy locations are monitored).

- Constructing (and maintaining) networks can be very costly, or networks can be faulty.



Cooper et al. 2021 [3]

Summary

- Sometimes, alternatives to the four-step transport model are useful
- Direct measurements (principle, examples, issues)
- Direct demand modelling (principle, examples, issues)
  - Interpolation and regression
  - Topological features
  - Space syntax
  - Built environment features
  - Data-based features

Further reading

1. Franke, T., et al. (2015). Smart crowds in smart cities: real life, city scale deployments of a smartphone based participatory crowd management platform. *Journal of Internet Services and Applications*, *6*, 1-19.
2. Wang, D. W. et al. (2019). A modified inverse distance weighting method for interpolation in open public places based on Wi-Fi probe data. *Journal of Advanced Transportation*, *2019*.
3. Cooper, C. H., et al. (2021). Using multiple hybrid spatial design network analysis to predict longitudinal effect of a major city centre redevelopment on pedestrian flows. *Transportation*, 48(2), 643-672.
4. Penn, A., et al. (1998). Configurational modelling of urban movement networks. *Environment and Planning B: planning and design*, *25*(1), 59-84.
5. Zaouche, M., et al. (2023). Bayesian spatio-temporal models for mapping urban pedestrian traffic. *Journal of Transport Geography,* 111, 103647.